

Introduction to Machine Learning

Lab 8: Mean-shift and Label Propagation

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1 Motivation

- Implement the mean-shift algorithm for data clustering
- When some labels are available, observe the diffusion process achieved by mean-shift.
- What if we implement mean-shift and label propagation with fixed kernels?

2 Tasks

Please read Lecture 9 carefully before doing this lab work.

1. **Mean-shift:** Given a set of data $\mathbf{X} \in \mathbb{R}^{N \times D}$, implement the mean-shift algorithm to find the means of the clusters, i.e., for the t -th iteration

$$m^{(t)}(\mathbf{x}_n) = f_{K_h}(m^{(t-1)}(\mathbf{X})) = \sum_{m=1}^N \frac{K_h(m^{(t-1)}(\mathbf{x}_n), m^{(t-1)}(\mathbf{x}_m)) m^{(t-1)}(\mathbf{x}_m)}{\sum_{i=1}^N K_h(m^{(t-1)}(\mathbf{x}_n), m^{(t-1)}(\mathbf{x}_i))} \quad (1)$$

2. **Label propagation:** When some data points are labeled by one-hot vectors, i.e., $\mathbf{Y} \in \{0, 1\}^{N \times K}$ and $\mathbf{Y} \mathbf{1}_K \leq \mathbf{1}_N$, where K is the number of clusters, we can estimate the labels for the unlabeled data via a **label propagation** algorithm. Essentially, this algorithm treats the observed one-hot vectors as the distribution of labels. It works a variant of the above mean-shift algorithm, in which the kernels are defined based on the mean of samples (i.e., $m(\mathbf{x})$'s), while the input and output are the distributions of labels (i.e., normalized $\mathbf{y}$'s). (Hint: It can also be viewed as a Nadaraya-Watson estimator)
3. In Task 1 and 2, the kernel matrices are time-varying. What if we use time-invariant kernel $K_h(\mathbf{x}_n, \mathbf{x}_m)$ instead when implementing the mean-shift and the label-propagation algorithm? Observe their differences and think about the reasons behind the phenomena.